MIL-T-45896(AR)
7 July 1977
SUPERSEDING
See Section 6

MILITARY SPECIFICATION

TRANSFORMER ASSEMBLY

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers requirements, quality assurance provisions, and the preparation for delivery criteria for one type of transformer entitled Transformer Assembly which is used in the Sprint warhead.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposals, form a part of this specification to the extent specified herein:

STANDARDS

MILITARY

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-202	- Test Methods for Electronic and Electrical Component Parts.
MIL-STD-1168	- Lot Numbering of Ammunition.
MIL-STD-1169	- Packaging, Packing and Marking for Shipment of Inert Ammunition Components.
MIL-STD-1460	- Soldering of Electrical Connections and Printed Wiring Assemblies, Procedure For.

DRAWINGS

ORDNANCE CORPS

9254805	- Shell, Encapsulating Transformer.
9254806	- Transformer Assembly.
EL-9254806	- Equipment List.

Beneficial comments (recommendations, additions, deletions), and any pertinent data which may be of use in improving this document, should be addressed to: Commander, US Army Armament Research and Development Command, ATTN: DRDAR-QAN-Q, Dover, New Jersey 07801, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426), appearing at the end of this document, or by letter.

FSC 1135

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PUBLICATIONS

PICATINNY ARSENAL

QAI-PA-1 - Quality Assurance Instructions for Reporting Data on Nuclear Weapons.

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Materials. The contractor shall furnish objective evidence (see 6.2) that the material complies with the specifications listed on the applicable drawing.

3.2 Assembly. The transformer shall comply with all requirements specified on the drawing (dwg) 9254806, with all requirements specified herein, and with the requirements of all applicable specifications to the extent specified on the drawing and herein.

3.3 First article testing. Requirements for the submission of first article samples by the contractor shall be as specified in the contract (see 4.2).

3.4 Coded serialization requirements (see 6.1.2). The contractor shall supply feedback data on punch cards for any item for which coding is required in section 3. The feedback data shall be in accordance with QAI-PA-1.

3.5 Circuit resistance. The circuit resistance shall comply with Table I when tested as specified in 4.4.1.

Table I

Circuit	Resistance (Ohms)	From Pin	To Pin	*Test Code (See 3.4 and Table II)
Primary	0.175 ± 0.015	1	2	06001_
Secondary	0.025 ± 0.005	3	4	06002_

*The last digit of the test code number is derived from the test condition and associated digit shown in Table II.

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Table II

Test Conditions	Last Digit
Ambient	0
Random Vibration (3.8.1)(4.4.4.1)	1
Temperature Shock (3.8.2)(4.4.4.2)	2

3.6 Insulation resistance. The insulation resistance between pins 1 and 3 of the transformer assembly shall not be less than 50 megohms when subjected to 500 ± 50 volts direct current (vdc). Testing shall be as specified in 4.4.2.

3.7 Operation. With a 1.8 +0, -0.2 ohm resistive load connected across the transformer secondary and with a rectangular wave, which is symmetrical about 0 volts reference level (see Figure 1) and which has an amplitude of 26 ± 0.5 volts peak to peak and the characteristics described in Table III (Primary) applied to the primary, the secondary voltage shall be a symmetrical rectangular wave having an amplitude of 10.5 ± 1 volt peak to peak and the characteristics described in Table III (Secondary) when tested as specified in 4.4.3. For wave form characteristics described in Table III, see Figure 1. For secondary voltage amplitude (peak to peak) use Test Code 08001-.

Table III

Characteristic	Primary	Secondary
1. Period	20 ± 1 microseconds	20 ± 1 microseconds
2. Rise Time	250 nanoseconds max	250 nanoseconds max
3. Fall Time	250 nanoseconds max	250 nanoseconds max
4. Pulse amplitude variations including overshoot, ringing, rounding of the leading or trailing edge and sag or droop	± 2.5 volts max	± 1.2 volts max

3.8 Environmental testing (see 4.4.4).

3.8.1 Random vibration. The completed transformer shall be able to withstand, without evidence of cracking, splitting, separation of parts or any other damage along its radial and vertical axes, the vibration environment given in Table IV. Attenuated at 24 Db/octaves. Test duration 60 seconds along each axis. Testing shall be as specified in 4.4.4.1.

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Table IV

Range (Hertz)	Power Level g^2/Hz
50-750	.1125
750-2500	.135
2500-3000	.45

(23.25 g's rms)

3.8.2 Temperature shock. The complete transformer shall withstand 3 cycles of temperature shock, without evidence of cracking, splitting, separation of parts or any other damage. A cycle shall consist of two 205°F changes in temperature, in the temperature range of -40° to 165°F as the temperature goes from hot to cold to hot when tested in accordance with paragraph 4.4.4.2. The transfer of the transformer between the two extreme temperature environments shall occur within a 5 minute interval. After completion of the temperature cycling, the transformer shall be returned to ambient room temperature without forced cooling and shall comply with the requirements of 3.5, 3.6 and 3.7.

3.9 Soldering. All soldering shall conform to MIL-STD-1460(MU).

3.10 Workmanship.

3.10.1 Plastic and metal characteristics. All components shall be free from cracks, splits, cold flow, shrinkage, cold shuts, inclusions, porosity, or any similar characteristic.

3.10.2 Threads. Threads shall be full and undamaged for the entire minimum length or depth as required on the applicable drawing.

3.10.3 Burrs. Parts shall be free from burrs.

3.10.4 Foreign matter. Parts shall be free from chips, dirt, grease, rust, corrosion, or any embedded foreign material.

3.10.5 Cleaning. The cleaning method shall not be injurious to any of the parts, nor shall the parts be contaminated by the cleaning agent.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved

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by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 First article inspection.

4.2.1 Quantity to be submitted. When the contract requires first article inspection, the contractor shall submit 6 complete assemblies (see dwg 9254806) for first article testing to the testing facility designated in the invitation for bids or request for proposal.

4.2.2 Inspection and testing to be performed. The first article inspection sample shall be subjected to the inspections and tests specified in Table V. Tests shall be conducted in order specified in Table V. The transformers that are not potted shall only be tested as specified in 4.3.2.1.1. The potted transformers shall be tested to the remaining tests.

Table V

First article inspection and tests

Inspection or Test	Dest	Non-Dest
1. Classification of characteristics (see 4.3.2.1)		X
2. Circuit resistance (see 3.5)		X
3. Insulation resistance (see 3.6)		X
4. Operation (see 3.7)		X
5. Random vibration (see 3.8.1)		X
6. Temperature shock (see 3.8.2)		X

4.2.3 Inspection equipment. The inspection and testing specified in 4.2.2 shall be accomplished by using the gages and test equipment specified in 4.3.4.1.

4.2.4 Rejection. If any item fails to comply with any of the applicable requirements, the first article sample shall be rejected.

4.3 Quality conformance inspection.

4.3.1 Lot formation.

4.3.1.1 Production lot. A lot shall consist of parts and assemblies produced by one manufacturer in one unchanged process, in accordance with the same drawing, same drawing revision, same specification and same specification revision. Drawing, specification and process changes not affecting safety, performance, storage, or interchangeability, as determined by the Government, shall not necessitate changing the lot interfix number. The criteria and procedures for the assignment of lot numbers shall be in accordance with MIL-STD-1168.

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4.3.1.2 Inspection lot. A lot or batch and its formation, size, and presentation is described in MIL-STD-105, Section 5. Accordingly, a lot shall mean an inspection lot and a batch shall mean an inspection batch for the purpose of this specification. The manner in which each inspection lot or batch is to be presented and identified by the contractor, shall be designated or approved by the Government representative.

4.3.2 Product inspection examination. A sample shall be selected at random from each inspection lot in accordance with the applicable major or minor defects/characteristics table (see Table VI and Table VII), and inspected for all defects/characteristics in the applicable classification. Any defects/characteristics in the sample found to be defective shall reject the inspection lot. Sequence for the inspection of defects/characteristics in each classification need not be followed as long as all defects/characteristics inspections are performed.

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Table VI
Major Defects/Characteristics

Lot Size	Sample Size	Lot Size	Sample Size	Lot Size	Sample Size	Lot Size	Sample Size
1-72	all	140-141	124	256-259	177	541-549	230
73	72	142	125	260-262	178	550-559	231
74	73	143-144	126	263-265	179	560-569	232
75	74	145	127	266-268	180	570-579	233
76	75	146-147	128	269-271	181	580-590	234
77	76	148-149	129	272-275	182	591-601	235
78	77	150	130	276-278	183	602-613	236
79	78	151-152	131	279-282	184	614-625	237
80	79	153-154	132	283-285	185	626-637	238
81-82	80	155-156	133	286-289	186	638-649	239
83	81	157	134	290-293	187	650-662	240
84	82	158-159	135	294-296	188	663-676	241
85	83	160-161	136	297-300	189	677-690	242
86	84	162-163	137	301-304	190	691-704	243
87	85	164-165	138	305-308	191	705-719	244
88-89	86	166-167	139	309-312	192	720-735	245
90	87	168	140	313-316	193	736-751	246
91	88	169-170	141	317-320	194	752-768	247
92	89	171-172	142	321-324	195	769-785	248
93	90	173-174	143	325-329	196	786-804	249
94-95	91	175-176	144	330-333	197	805-823	250
96	92	177-178	145	334-338	198	824-842	251
97	93	179-180	146	339-342	199	843-863	252
98	94	181-182	147	343-347	200	864-884	253
99-100	95	183-184	148	348-352	201	885-907	254
101	96	185-187	149	353-356	202	908-930	255
102	97	188-189	150	357-361	203	931-955	256
103	98	190-191	151	362-366	204	956-981	257
104-105	99	192-193	152	367-372	205	982-1008	258
106	100	194-195	153	373-377	206	1009-1036	259
107	101	196-197	154	378-382	207	1037-1066	260
108-109	102	198-200	155	383-388	208	1067-1098	261
110	103	201-202	156	389-393	209	1099-1131	262
111	104	203-204	157	394-399	210	1132-1166	263
112-113	105	205-207	158	400-405	211	1167-1204	264
114	106	208-209	159	406-411	212	1205-1243	265
115	107	210-212	160	412-417	213	1244-1285	266
116-117	108	213-214	161	418-424	214	1286-1330	267
118	109	215-217	162	425-430	215	1331-1378	268
119	110	218-219	163	431-437	216	1379-1429	269
120-121	111	220-222	164	438-443	217	1430-1483	270
122	112	223-224	165	444-450	218	1484-1542	271
123-124	113	225-227	166	451-457	219	1543-1605	272
125	114	228-230	167	458-465	220	1606-1673	273
126-127	115	231-232	168	466-472	221	1674-1747	274
128	116	233-235	169	473-480	222	1748-1827	275
129-130	117	236-238	170	481-488	223	1828-1914	276
131	118	239-241	171	489-496	224	1915-2010	277
132-133	119	242-243	172	497-504	225	2011-2115	278
134	120	244-246	173	505-513	226	2116-2232	279
135-136	121	247-249	174	514-521	227	2233-2361	280
137	122	250-252	175	522-530	228	over 2361	298
138-139	123	253-255	176	531-540	229		

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Table VII
Minor Defects/Characteristics

Lot Size	Sample Size	Lot Size	Sample Size	Lot Size	Sample Size	Lot Size	Sample Size
1-16	all	28-29	23	52-56	31	132-156	39
17	16	30-31	24	57-62	32	157-191	40
18	17	32-34	25	63-69	33	192-244	41
19-20	18	35-36	26	70-76	34	245-334	42
21	19	37-40	27	77-86	35	335-519	43
22-23	20	41-43	28	87-97	36	520-1120	44
24-25	21	44-47	29	98-112	37	over 1120	45
26-27	22	48-51	30	113-131	38		

4.3.2.1 Classification of defects/characteristics (see 3.2). Critical, major and minor defects/characteristics are defined in MIL-STD-105. Inspection shall be conducted as follows:

4.3.2.1.1 Transformer assembly (before potting) (see dwg 9254806).

Categories	Defects/Characteristics	Method of Inspection
Critical:	None defined.	
Major:		
	101. Objective evidence that materials are as specified (see 3.1)	Visual
	102. Any component cracked, damaged, loose or missing	Visual
	103. Soldering improper (see 3.9)	Visual
Minor:		
	201. Evidence of poor workmanship (see 3.10)	Visual

4.3.2.1.2 Transformer assembly (after potting) (see dwg 9254806).

Categories	Defects/Characteristics	Method of Inspection
Critical:	None defined.	
Major:		
	101. Objective evidence that materials are as specified (see 3.1)	Visual

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102. Height of potting incorrect.....Gage
103. Pin location incorrect (see dwg 9254805).....Gage
104. Pin height incorrect (see dwg 9254805).....Gage
105. Shell outside diameter incorrect (see dwg 9254805).....Gage
106. Marking improper, missing or illegible.....Visual
107. Objective evidence that all electrical results have been recorded (see 3.4).....Visual
108. Any pin bent or damaged.....Visual

Minor:
201. Evidence of poor workmanship (see 3.10).....Visual

4.3.2.1.3 Preparation for delivery (see 5).

Categories	Defects/Characteristics	Method of Inspection
Critical:	None defined.	
Major:		
	101. Preservation, packing, and packaging not in accordance with MIL-STD-1169	Visual
	102. Marking not in accordance with MIL-STD-1169	Visual
Minor:		
	201. Evidence of poor workmanship (see 3.10)	Visual

4.3.3 Testing. All tests shall be performed. Tests shall be performed in the order specified below:

Test	Test Classification
a. Circuit resistance	Major
b. Insulation resistance	Major
c. Operation	Major

None of these tests are considered destructive and samples so tested may be returned to the lot.

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4.3.4 Acceptance inspection equipment.

4.3.4.1 Major defects/characteristics and tests. Inspection and testing shall be performed with the acceptance inspection equipment, operating instructions, and calibration instructions specified on the Equipment Lists (EL) (see EL-9254806). When either of the conditions listed below exist, the contractor shall design required equipment or specify suitable commercial equipment and obtain approval prior to use on the contract.

a. Any deviation from equipment or procedures specified on the EL is desired (see 6.3).
b. The code "CD" is listed in the "Part or Identifying Number" column on the EL (see 6.1.2).

4.3.4.2 Minor defects/characteristics and tests. Acceptance inspection equipment for minor defects/characteristics is not specified on ELs. The acceptance inspection equipment, operating instructions and calibration instructions used by the contractor for minor classification inspection shall be approved by the Government representative authorized by the procuring agency.

4.4 Test methods. The tolerances specified herein are absolute with no allowance for test equipment inaccuracy. The tolerances used by the manufacturer shall be equal to the absolute tolerances less the accuracy tolerances of the test equipment used.

4.4.1 Circuit resistance. The circuit resistance shall be measured in accordance with Table I (see 3.5) with approved measuring equipment.

4.4.2 Insulation resistance. The insulation resistance of the transformer assembly shall be measured with an approved insulation resistance tester in accordance with MIL-STD-202, Method 302, Test Condition B for 10 seconds minimum or until meter stabilizes. The test current shall be limited to 10 milliamperes.

4.4.3 Operation. The operation of the transformer assembly shall provide the output specified in 3.7 when tested with approved test equipment. Permanent records of output values and wave form shall be made of each test.

4.4.4 Environmental testing (see 3.8).

4.4.4.1 Random vibration. The transformer assembly shall be rigidly mounted to a fixture capable of being supported readily in each of three mutually perpendicular axes. The unit shall be vibrated over the frequency range specified in 3.8.1. After this test, the assembly shall meet the requirements of 3.5, 3.6 and 3.7.

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4.4.4.2 Temperature shock. The transformer assembly shall be placed in a test chamber that has been stabilized (see 6.4) at 165 + 5°F and maintained for a minimum of 4 hours. The unit shall then be removed from the high temperature chamber and placed in a chamber having internal temperature stabilized (see 6.4) at -40 + 5°F within 5 minutes, and maintained at this temperature for a minumum of 4 hours. Within 5 minutes the transformer assembly shall be returned to the high temperature chamber that is stabilized at 165 + 5°F. This sequence constitutes one cycle. Three cycles shall be executed. Upon completion of the third cycle, return the unit to ambient room temperature without forced cooling. After this test, the assembly shall meet the requirements of 3.5, 3.6 and 3.7.

5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging. Preservation and packaging shall be level A or C as specified (see 6.1).

5.1.1 Level A. The transformer shall be preserved and packaged in accordance with instructions contained in MIL-STD-1169(PA).

5.1.2 Level C. The transformer shall be preserved and packaged in accordance with instructions contained in MIL-STD-1169(PA).

5.2 Packing.

5.2.1 Levels A and C. The transformer shall be packed in accordance with the applicable instructions contained in MIL-STD-1169(PA).

5.3 Marking. Marking shall be in accordance with MIL-STD-1169(PA).

6. NOTES

6.1 Ordering data. Procurement documents should specify the following:

6.1.1 Procurement requirements.

a. Title, number, and date of this specification.

b. Selection of applicable level of packaging and packing required (see 5.1 and 5.2).

6.1.2 Contract data requirements (see 3.5 and 4.3.4.1). Data specified in 3.4 and 4.3.4.1 will be delivered as identified on a numbered DD Form 1664 when required on a DD Form 1423 incorporated in the contract.

6.2 Objective evidence. Records of contractor quality control and inspections which can be verified.

6.3 Acceptance inspection equipment (see 4.3.4.1). The contractor shall obtain approval from the Commander, US Army Armament Research and Development Command, ATTN: DRDAR-QAN-I, Dover, New Jersey 07801.

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6.4 Temperature stabilization. The temperature is considered stabilized when the greatest peaks (both highest and lowest) continually observed over 5 minutes min. do not exceed the nominal temperature plus tolerance specified.

6.5 Supersession data. This specification supersedes the requirements of Purchase Description PA-PD-2239 dated 29 November 1972 and SCN No. 1 dated 4 March 1974.

Custodian:
Army - (AR)

Preparing activity:
Army - (AR)

Project No. 1135-A218

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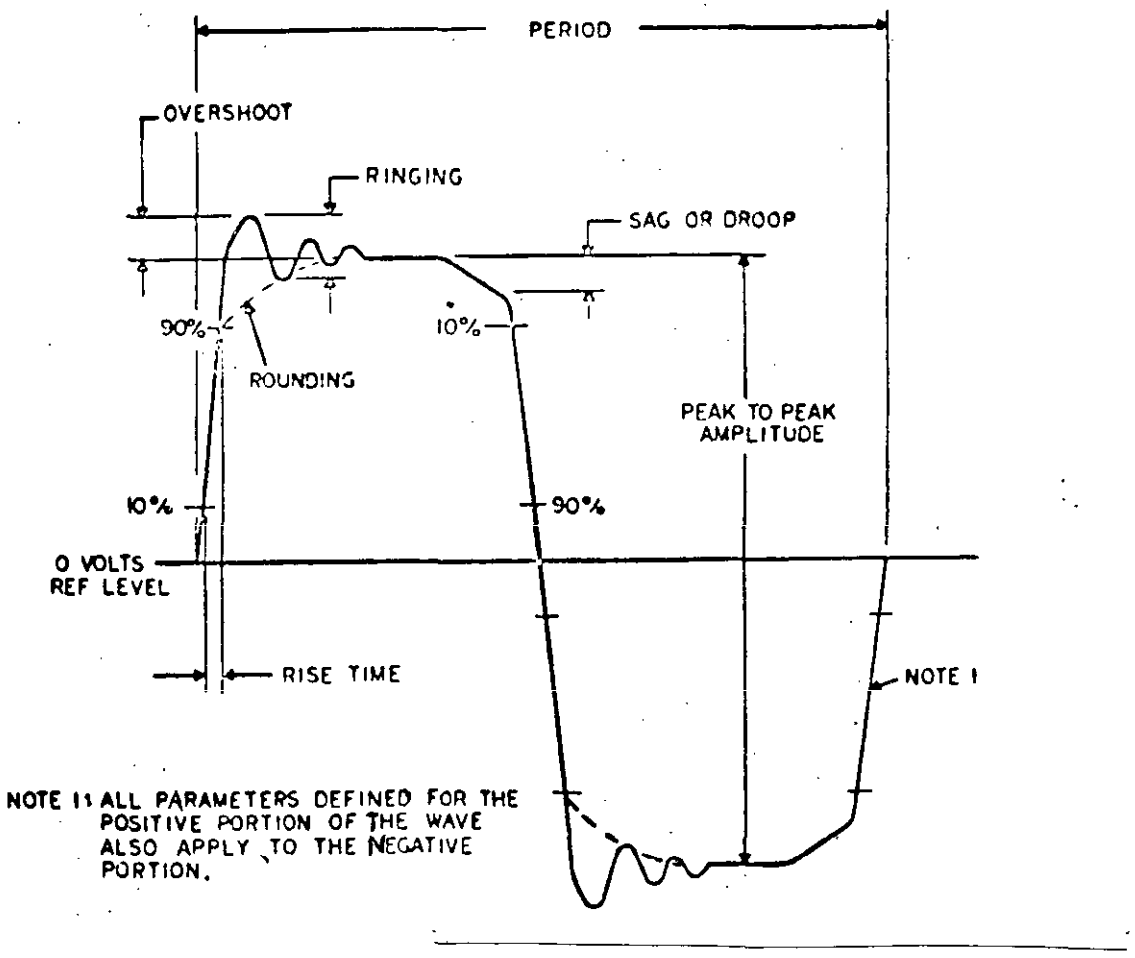


Figure 1.

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

OMB Approval
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DOCUMENT IDENTIFIER AND TITLE

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

[ ] DIRECT GOVERNMENT CONTRACT [ ] SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3. IS THE DOCUMENT RESTRICTIVE?

YES NO *(If "Yes", in what way?)*

4. REMARKS

SUBMITTED BY *(Printed or typed name and address - Optional)*

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DATE

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